

Supplemental Information for

Effective Network Profiles of the Human Cingulate Gyrus

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Supplementary Text

Supplemental Results

Temporal Dynamics of the Effective Network of Cg Subregions

We implemented a changepoint detection algorithm to determine the average onset/offset latency and duration of CCEPs across regions for each stimulation condition (Supplemental Figure 7a & 7b). Supplemental Figure 7c shows the normalized response latency across brain regions in each stimulation condition. CCEPs in the ACC and MCC conditions showed homogeneity in onset time, with higher latency responses occurring in the hippocampus and at the temporal poles. The ACC also showed higher latency responses in the postcentral sulcus, the inferior temporal lobe, and the supramarginal gyrus. The PCC condition showed a wider spread of response onset latencies, with later responses occurring in the central and superior insula, frontal lobe, inferior and posterior temporal lobe, and parietal lobe. We found a significant positive correlation between response latency and distance from the recording electrode to the stimulating electrodes (Supplemental Figure 8; Spearman's $\rho = 0.33$, $p < 0.0001$). Response duration showed a similar comparative trend between regions, with the most variable spread of high-duration CCEPs in the PCC stimulation condition (Figure 7d). The ACC condition had the longest responses in the hippocampus, postcentral sulcus, superior temporal lobe, and precuneus. The MCC condition had the longest responses in the parahippocampal gyrus, inferior frontal gyrus, and supramarginal gyrus. The PCC condition showed high duration responses throughout the frontal lobe, hippocampus, temporal pole, and short insular gyri.

Cg Stimulation-Evoked Gamma Responses across the Cortex

Beyond modeling Cg effective connectivity through SPES-induced CCEPs, we also evaluated evoked broadband gamma to characterize average changes in local neuronal firing within each region. Supplemental Figure 9a shows the average evoked broadband gamma profiles for each stimulation condition calculated from contacts showing both significant effective connectivity and significant changes in broadband gamma. Most regions showed a SPES-induced increase in broadband gamma in the ACC and MCC conditions. The greatest increase in broadband gamma power was found in the hippocampus for the ACC condition, and the middle frontal gyrus in the MCC condition. The PCC evoked broadband gamma response profile was heterogeneous, showing increases and decreases in broadband gamma power across the cortex. Notable increases in broadband gamma occurred at the temporal pole, ACC, posterior MCC, and the superior segment of the circular sulcus of the insula. Notable decreases were observed in the superior frontal gyrus, the hippocampal body, the short insular gyri, and the anterior segment of the circular sulcus of the insula. Supplemental Figure 9b shows the proportion of contacts determined to be significantly effectively connected to the Cg subregions that also showed a significant SPES-induced change in broadband gamma (64.3% for ACC, 66.8% for MCC, 55.7% for PCC).

Supplemental Figure 10 compares the evoked gamma responses observed across all conditions and between hemispheres. There were significant differences between all three conditions ($p = 0.003$ for ACC vs MCC, $p = 7.12\text{E-}9$ for MCC vs PCC, $p = 6.33\text{E-}18$ for ACC vs PCC). There were also significant hemispheric differences observed between left and right Cg stimulation for the MCC ($p = 4.86\text{E-}29$) and the PCC ($p = 0.006$). Supplemental Table 2 lists the statistical information comparing the ipsilateral evoked gamma in response to stimulation of left or right Cg subregions. Notably, there was a significant difference in evoked gamma between the left and right frontal lobes for the ACC condition ($p = 0.005$), frontal lobes, insula, orbitofrontal cortex, and temporal lobes for the MCC condition ($p = 8.38\text{E-}7$, $p = 5.61\text{E-}5$, $p = 5.59\text{E-}8$, $p = 0.004$, respectively), and the frontal lobes for the PCC condition ($p = 3.39\text{E-}5$). Several regions also demonstrated a difference in gamma response direction (increased or decreased gamma) between hemispheres (Supplemental Table 3). For the MCC condition, gamma responses increased in the left orbitofrontal cortex ($\text{Rho} = 0.22$) and decreased in the right ($\text{Rho} = -0.07$). For

the PCC condition, there was an increase in the left frontal lobe (0.06) and a decrease in the right (-0.28).

Supplemental Methods

Automated Dendrogram Distance Threshold Selection

The algorithm evaluated clustering solutions across distance thresholds (τ) from 0.1 to 0.9 with a $\Delta\tau$ of 0.01 using a composite scoring function that combined natural breaks in dendrogram structure, balance in cluster sizes, and Shannon entropy. Before scoring, for each τ , any clusters containing two or fewer regions were not considered and removed from the analysis. Breakpoint detection was calculated by finding the merge densities across the range of merge heights from the linkage matrix Z :

$$M(h) = \frac{Hist.(Z_{:,3}, n_{bins})}{\max(Hist.(Z_{:,3}, n_{bins}))}$$

Equation 2. Calculation of merge densities across dendrogram heights.

Where M is the normalized merge density as a function of dendrogram height h , and $Hist.$ represents the histogram function with $n_{bins} = 20$. Local minima representing natural breakpoints were identified by detecting sign changes in the first derivative of the function M . The score for the natural break points for each τ was calculated as:

$$S_{breaks}(\tau) = 1 - \frac{\min(|\mathcal{M} - \tau|)}{\max(Z_{:,3})}$$

Equation 3. Calculation for the breakpoint score to represent naturally occurring gaps in tree structure at each τ .

To calculate a score to represent the similarity of cluster sizes, we calculated a coefficient of variation CV at each τ as the ratio of the standard deviation in the number of regions per cluster to the average number of clusters (where τ' represents all threshold values). We then calculated a cluster balance score as:

$$S_{balance}(\tau) = 1 - \frac{CV(\tau) - \min_{\tau'}(CV(\tau'))}{\max_{\tau'}(CV(\tau')) - \min_{\tau'}(CV(\tau')) + \epsilon}$$

Equation 4. Calculation for cluster balance score as a metric to represent similarity in cluster size and variance at each τ .

Where $\epsilon = 10E-16$ to prevent division by zero. Shannon entropy was calculated as:

$$E(\tau) = - \sum_{i=1}^k p_i \log_2(p_i + \epsilon)$$

Equation 5. Calculation for Shannon entropy at each τ , representing the diversity of cluster sizes.

Where $p_i = \frac{|c_i|}{\sum_{j=1}^k |c_j|}$ represents the proportion of regions in cluster c_i . The Shannon entropy score was then calculated as:

$$S_{entropy}(\tau) = \frac{E(\tau) - \min_{\tau'}(E(\tau'))}{\max_{\tau'}(E(\tau')) - \min_{\tau'}(E(\tau')) + \epsilon}$$

Equation 6. Calculation for Shannon entropy at each τ , representing the distribution of regions across clusters.

To account for the number of valid regions included in the clusters and the overall distribution of tree structure at each τ we used a composite weight calculated from the following:

$$w_{\text{adaptive}}(\tau) = \exp\left(-\frac{(r(\tau) - 0.5)^2}{0.15}\right)$$

Equation 7. Adaptive weight to account for the distribution of merge densities above and below τ , where $r(\tau)$ represents the proportion of merges below τ .

$$I_{\text{retention}} = 0.5 + 0.5 \times (1 - \exp(-\sigma_{\text{tree}}))$$

Equation 8. Asymptotic function that prioritizes region retention when dendrogram merge heights are unevenly spaced, avoiding thresholds that would exclude large proportions of regions.

$$\sigma_{\text{tree}} = \frac{\text{std}\left(\text{diff}\left(\text{sort}(Z_{:,3})\right)\right)}{\text{mean}\left(\text{sort}(Z_{:,3})\right)}$$

Equation 9. Implementation of σ_{tree} : a metric measuring variability in spacing between clustering events in the dendrogram. Higher values represent larger variability in spacing between successive merge heights, and lower scores show evenly spaced merge heights. At higher values, breakpoints in the tree structure are more likely to exclude a larger portion of regions. Here, $\text{diff}\left(\text{sort}(Z_{:,3})\right)$ represents the successive differences of merge heights sorted in ascending order.

$$w_{\text{composite}}(\tau) = w_{\text{adaptive}}(\tau) \times \left(\frac{n_{\text{regions}}(\tau)}{n_{\text{total}}}\right)^{I_{\text{retention}}}$$

Equation 10. Composite weight function incorporating adaptive emphasis on regional retention (i.e., minimizing the number of regions removed from the final clustering structure) and preferential selection of τ values near the median tree height.

The composite optimization function was then calculated as:

$$F(\tau) = w_{\text{composite}}(\tau) \times [2.0 \times S_{\text{breaks}}(\tau) + 1.5 \times S_{\text{balance}}(\tau) + 1.5 \times S_{\text{entropy}}(\tau)]$$

Equation 11. Composite optimization function for each τ that takes into account the breakpoint structure of the dendrogram, the number of regions represented across clusters, and the Shannon entropy of clusters.

The optimal threshold was then calculated as $\arg \max_{\tau} F(\tau)$, such that the resulting number of valid clusters (>2 regions) was greater than 1.

Temporal Dynamics of Cg Effective Connectivity Networks

To characterize the onset latency and duration of CCEPs across stimulation conditions, we applied an optimal detection of changepoints algorithm on the mean absolute value of the Hilbert transformed broadband CCEPs (broadband magnitude), using two changepoints (under the assumption that a CCEP contains a start and end) at each contact. This method identified changepoints in the time series such that the statistical properties of data between changepoints were maximally separated from the statistical properties of data outside of the changepoints (1). For the onset detection of CCEPs (or CCEP latency), we applied this method to the first derivative of the broadband magnitude to identify the onset and offset of rate changes in the LFP. For offset detection of the CCEPs, we used the changepoint detection method on the broadband

magnitude and used the second identified changepoint. The duration of the response was calculated as the difference between the offset and onset.

To ensure that changepoints were detected within expected onset/offset ranges, we estimated the spectral smearing introduced to the signal through broadband filtering, then excluded any changepoints detected outside of the spectral smearing range. Spectral smearing was estimated by applying the broadband filter to simulated data containing a single square pulse, then measuring the change in pulse onset before and after filtering. We found that the degree of smearing depended on various factors, including the magnitude and slope of the evoked responses, so we were unable to extract accurate onset times of stimulation; however, through normalization, we were able to capture trends in response onset between regions in each stimulation condition.

Evoked Gamma Responses to SPES

We characterized the evoked broadband gamma responses across regions using the extracted broadband gamma. To characterize increases or decreases in broadband gamma power at each contact, we used a rank-biserial Spearman's correlation between the mean absolute value of the Hilbert transformed broadband gamma signal at each contact and a dichotomy (i.e., a generated data vector of zeros and ones corresponding to the lengths of the baseline and CCEP windows for each signal) (2). We then took the average Spearman's rho across all contacts within each region to represent the change in broadband gamma across regions for each stimulation condition.

Extracting Angular Features of CCEPs

Using the defined onset and offset (CCEP response window) indices we derived trial-wise instantaneous phase from the complex broadband signal derived from applying a Hilbert transformation. To increase cross-trial phase consistency and suppress isosyncratic phase excursions, we applied a normalization to align each trial's phase to the across-trial mean at each time point such that $\tilde{\phi}_i(t) = \cos(\phi_i(t) - \bar{\phi}(t)) \bar{\phi}(t)$.

We extracted angular features using a peak-valley characterization constrained to the response window. A single-pass window tracked alternating local maxima and minima of the broadband signal's direction. Each directional changepoint was stored and ranked by absolute amplitude, and the three with the greatest magnitude were maintained and labeled Angle 1, Angle 2, and Angle 3 of the CCEP. Further, we stored the angle of the response onset, the onset of each angle, and the angle of the response offset.

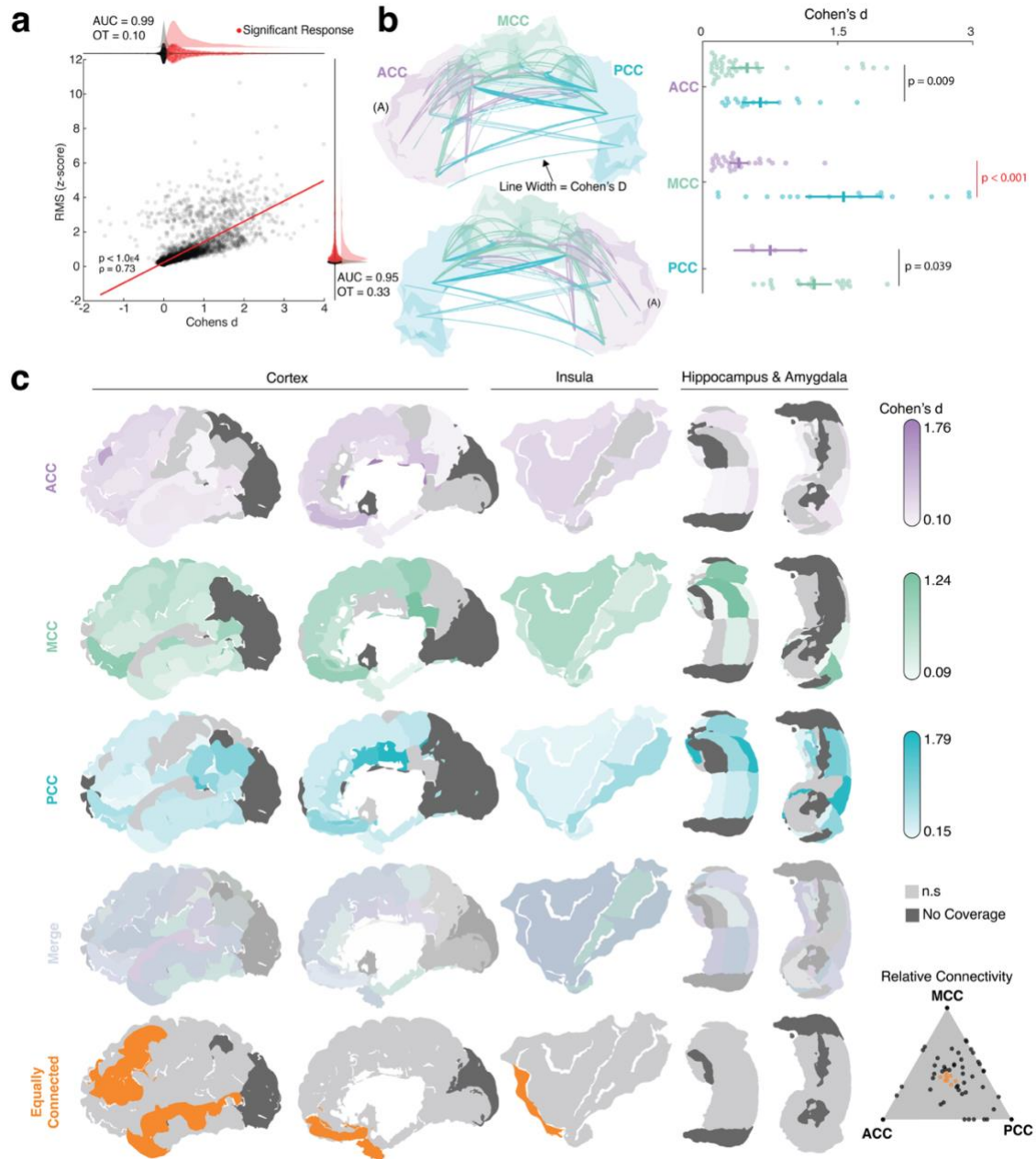
Extracting N1/N2 Features of CCEPs

We extracted CCEP peaks from trial-averaged waveforms using fixed post-SPES windows: 10-50ms for N1 and 60-700ms for N2. A peak was considered present only if its absolute height exceeded a threshold of 5 times the standard deviation of the pre-SPES baseline window. Peaks were found by locating discrete local maxima within the response window. The width of peaks was measured at half-prominence. Prominence was computed from the vertical distance to the lowest contour that connects the peak to higher neighbors.

Supporting Figures and Tables

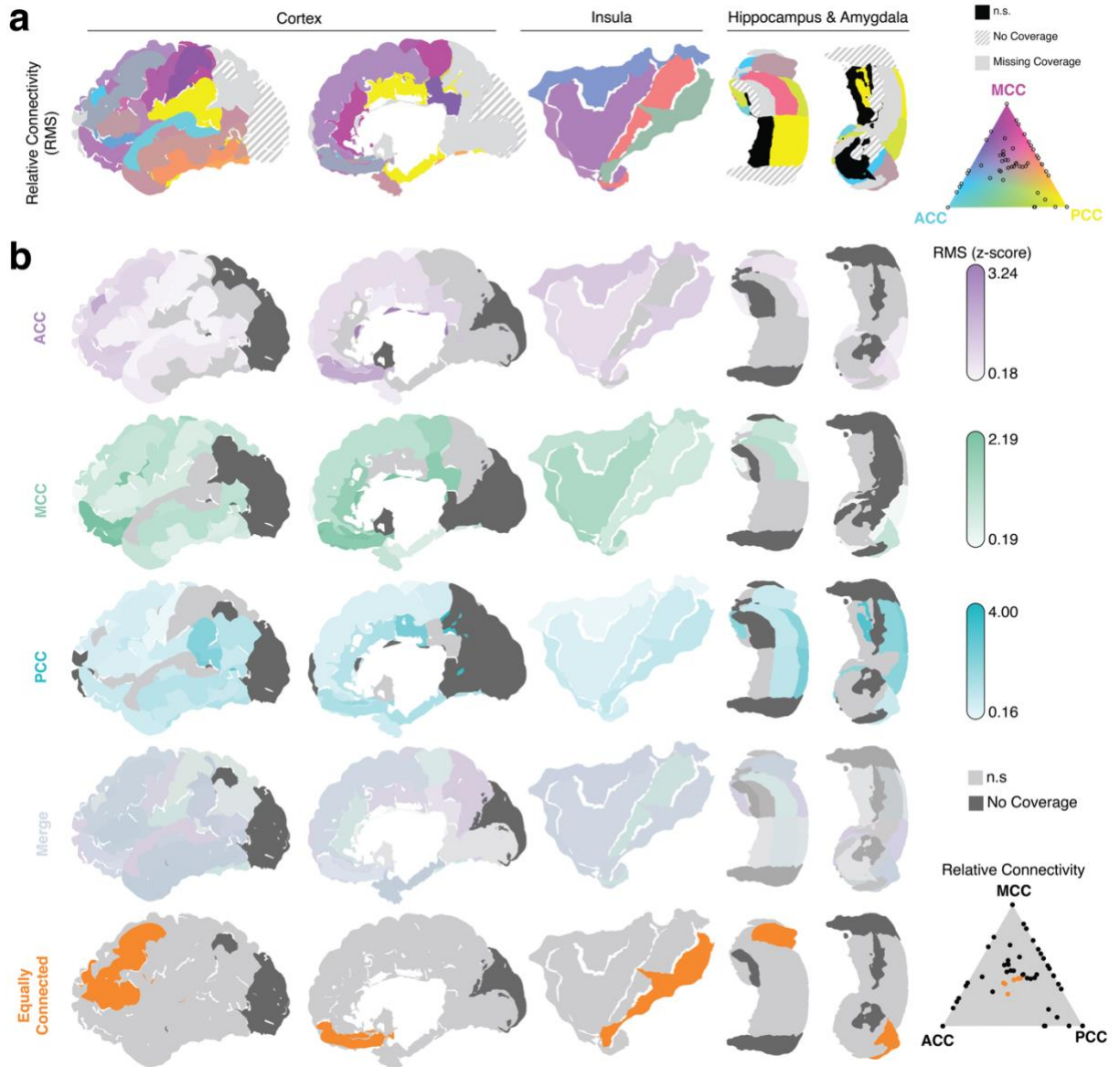


Supplemental Figure 1. Percentage of significant contacts across brain regions. Cortical surface models showing the percentage of significant contacts within each region across all three stimulation conditions (stimulation of the anterior, mid, and posterior cingulate cortices) and both measures of effective connectivity (Cohen's d and RMS).



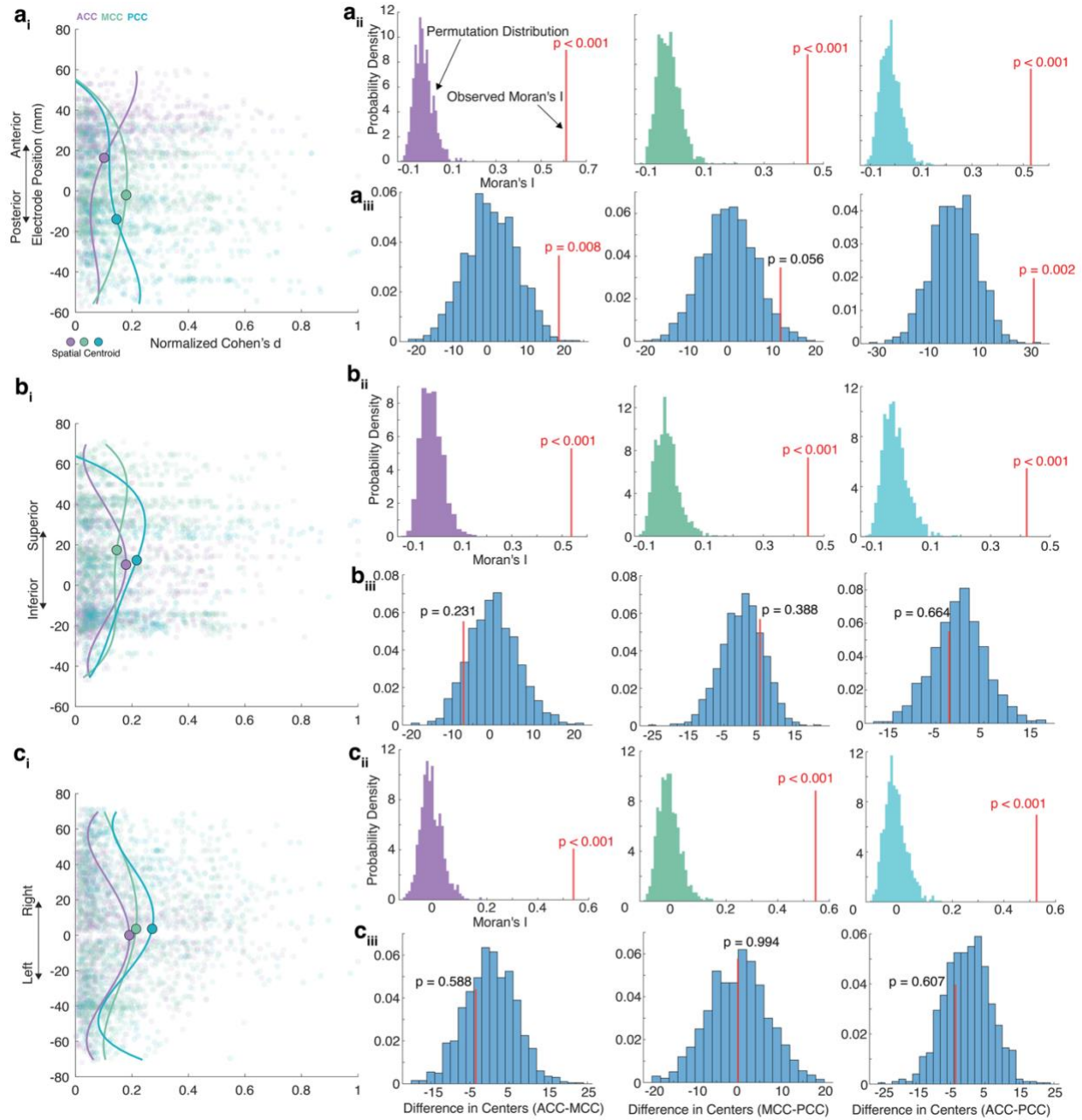
Supplemental Figure 2. Expanded cingulate cortex (CC) effective connectivity modeled using cortico-cortical evoked potential (CCEP) coherence. (a) Relationship between CCEP coherence (Cohen's d) and CCEP amplitude (RMS). The scatterplot shows the relationship between these two measures across all contacts, across all conditions. Distributions of each respective effective connectivity measure projected to the top (Cohen's d) and the right (RMS) axes show significantly connected channels (with respect to each measure) in red. The area under the curve of the receiver operating characteristic was used to determine the performance of classifying significantly effectively connected contacts for each measure. **(b)** Inter-CC subregion effective connectivity using CCEP coherence (Cohen's d) obtained from patients implanted with contacts in more than one CC subregion ($n = 5$ for ACC, $n = 6$ for MCC, $n = 5$ for PCC). (Left) Surface model of the CC showing connections between its subregions. Lines were drawn from stimulated contact coordinates to all significantly effectively connected contact coordinates in the other CC subregion(s). Colors of lines represent the stimulated subregion, and line width represents the normalized Cohen's d . (Right) Swarm plot comparing the effective connectivity of each CC subregion with each of the other two subregions. For each stimulation condition, the Cohen's d of contacts located within each of the other non-stimulated subregions was compared using a Wilcoxon rank sum test ($\alpha = 0.01$). Centrality lines show the

mean, and the whiskers show the 95% confidence interval. **(c)** (Rows 1-3) Cortical surface models showing effective connectivity, as average Cohen's d , across brain regions for each stimulation condition. (Row 4) Overlay of rows 1-3. (Row 5) Brain regions with highly similar effective connectivity (Cohen's d) to all three CC subregions, determined as any brain regions located within a circle with an area 5% of the triangular plane used to determine relative connectivity to the CC subregions.

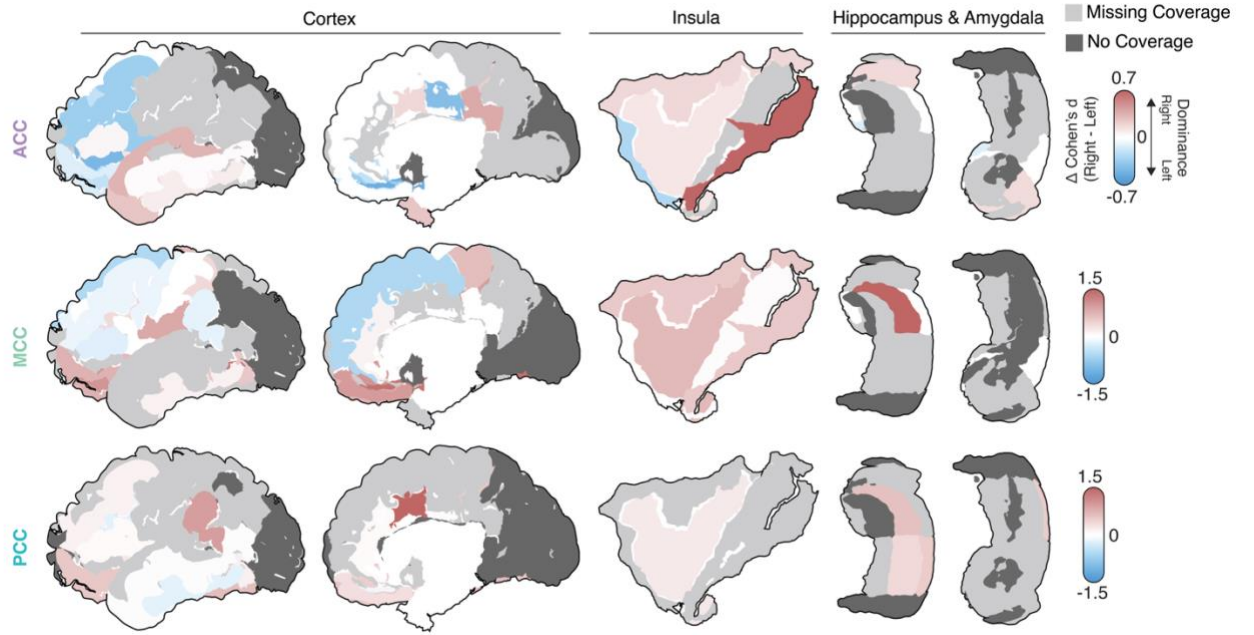


Supplemental Figure 3. Expanded cingulate cortex (CC) effective connectivity modeled using cortico-cortical evoked potential (CCEP) magnitude. (a) Cortical surfaces visualizing the relative effective connectivity, as measured by CCEP RMS, of each CC subregion to each cortical region (See Figure 2 and Methods). **(b)**

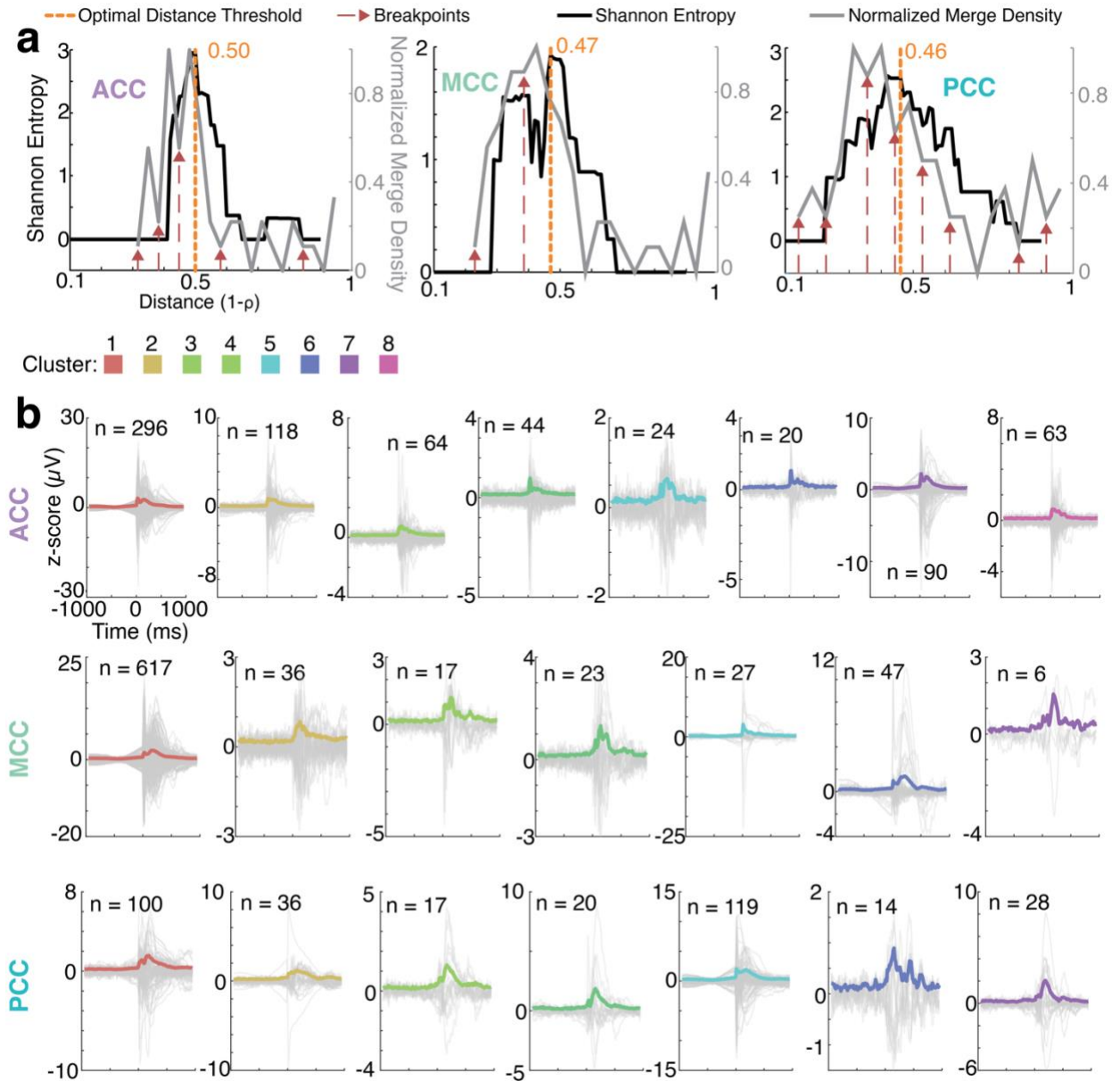
(Rows 1-3) Cortical surface models showing effective connectivity, as average CCEP RMS, across brain regions for each stimulation condition. (Row 4) Overlay of rows 1-3. (Row 5) Brain regions with highly similar effective connectivity (RMS) to all three CC subregions, determined as any brain regions located within a circle with an area 5% of the triangular plane used to determine relative connectivity to the CC subregions.



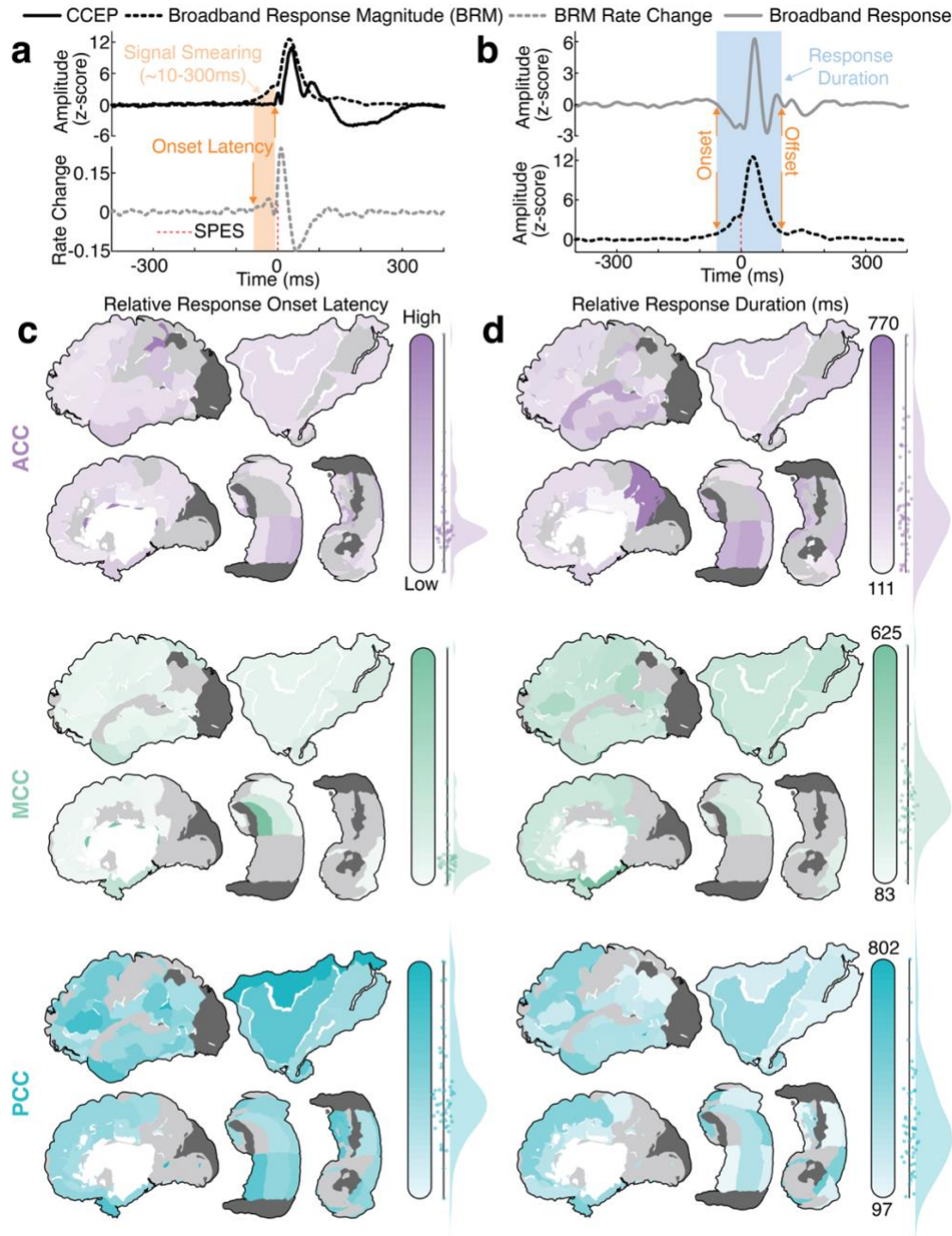
Supplemental Figure 4. Statistics for analyzing the spatial distribution of effective connectivity across the cortex. (a*i*) Distributions of normalized effective connectivity (Cohen's *d*) across the anterior-posterior axis. Lines show a 4th-order polynomial trend line fitted to the mean Cohen's *d* of 50 evenly spaced bins for each spatial dimension. The circles along the lines represent the centroids of each connectivity distribution. (a*ii*) Permutation distributions used to determine whether Moran's *I* for each stimulation condition was significant. (a*iii*) Permutation-based pairwise comparisons of observed centroid differences were calculated to compare the centroids of distributions for each stimulation condition. (b*i*-b*iii*) Same analysis as in **a** for the inferior-superior axis. (b*i*-b*iii*) Same analysis as in **a** for the left-right axis.



Supplemental Figure 5. Hemispheric differences in effective connectivity across cortical regions. Cortical surface models showing the difference in average Cohen's *d* between the left and right hemispheres across regions for each stimulation condition. Differences were calculated as average right hemisphere Cohen's *d* minus average left hemisphere Cohen's *d* for each brain region. Blue represents regions with greater Cohen's *d* in the right hemisphere, and red represents greater Cohen's *d* in the left hemisphere.

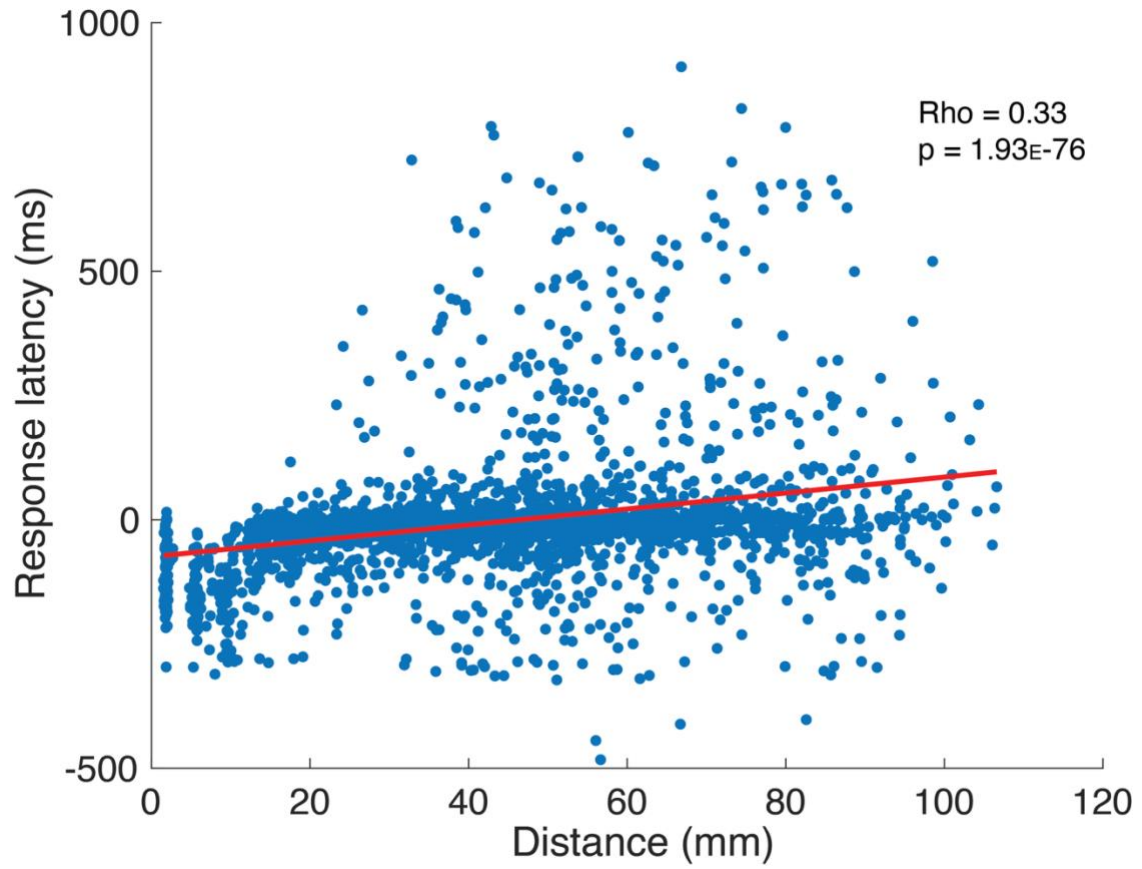


Supplemental Figure 6. Hierarchical clustering of cingulate cortex (CC) effective connectivity networks. (a) Optimal distance threshold selection for hierarchical clustering of each CC subregion's effective connectivity network dendrogram. Optimal distance to resolve clusters was calculated using a weighted combination of the Shannon Entropy (left axis) and normalized merge density (right axis) of network dendrograms. Plots show these metrics across all distances of each dendrogram. **(b)** Visualization of cortico-cortical evoked potential (CCEP) morphology across clusters for each stimulation condition. Colors correspond to each cluster. Grey lines show average CCEPs recorded at each contact within each cluster. Colored lines show the average magnitude of the CCEPs within each cluster (to account for differences in CCEP polarity across contacts).

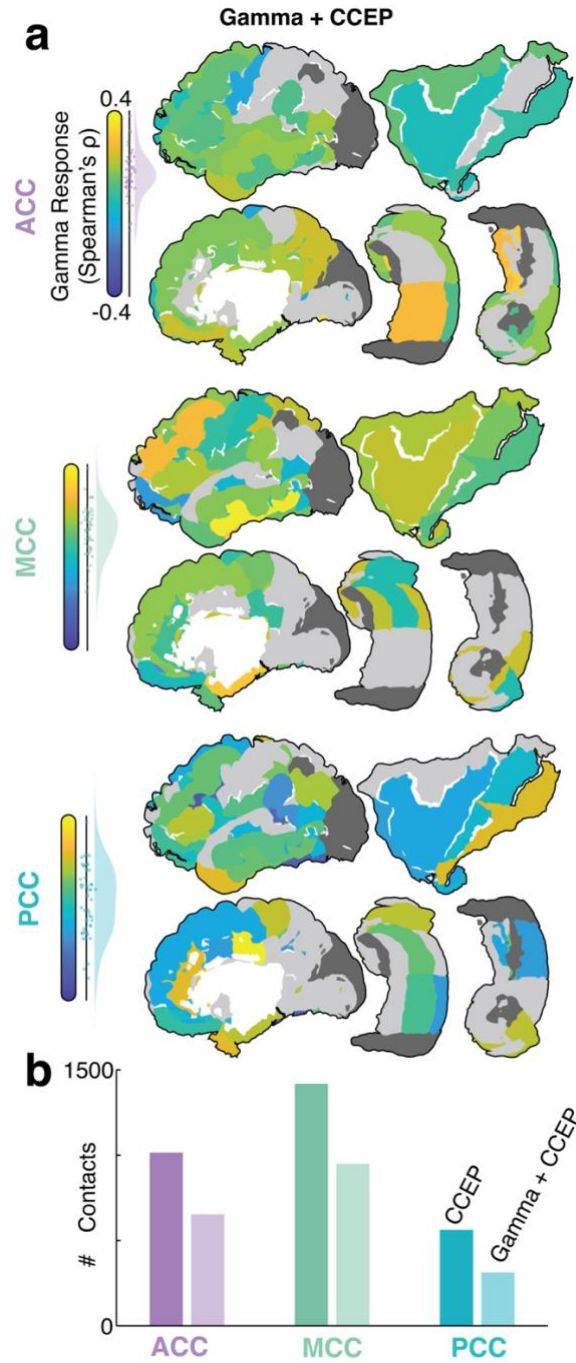


Supplemental Figure 7. *Temporal Dynamics of Cingulate Cortex Effective Connectivity.* **(a)** Cortico-cortical evoked potential (CCEP) onset was detected using a time-series optimal partitioning algorithm (see Methods) to identify the first instance relative to stimulation, where the statistical properties of the broadband (5–40 Hz) local field potential (LFP) slope changes from baseline. For this analysis, the temporal distortion caused by bandpass filtering created a spectral smearing artifact, resulting in the detected onset of the CCEP to occur prior to stimulation in many cases. To account for this, the onset was normalized across conditions, enabling comparison of response onset latencies between regions, at the cost of accurate onset values. **(b)** Response duration was calculated using both the response onset and the response offset. The response offset was calculated using the optimal partitioning method on the broadband CCEP response magnitude, specifically, by identifying the two indices that best separate the statistical properties of the broadband CCEP. The second of these two indices was used as the CCEP offset. **(c)** Anatomical visualization of normalized response onset latencies across regions for each stimulation condition. Darker colors indicate later CCEP onsets. Distributions of onset latencies across regions are shown adjacent to the

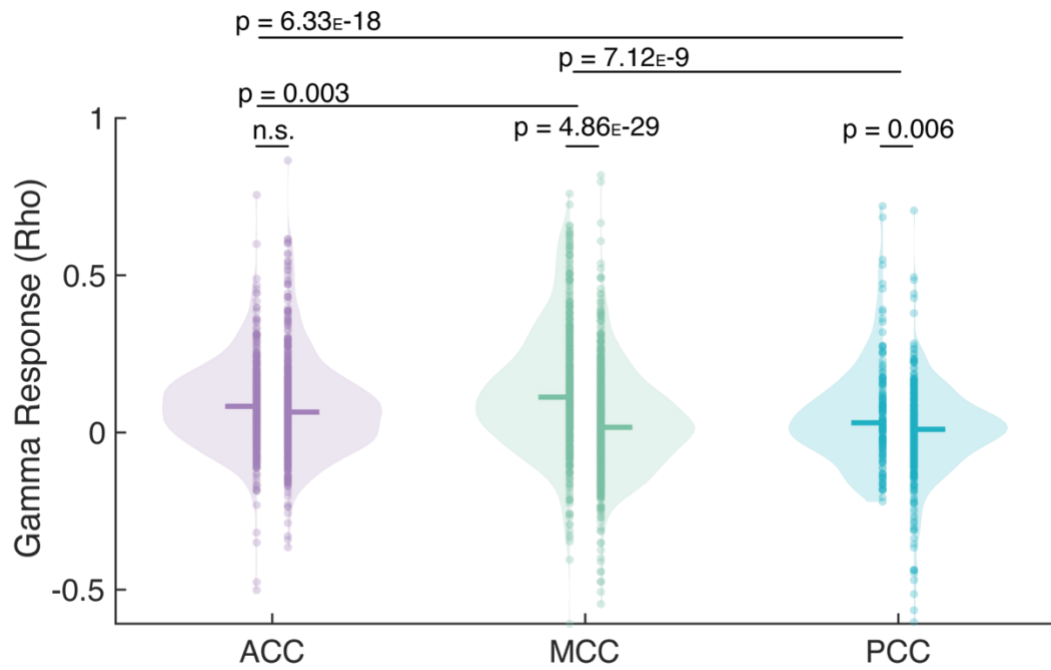
colorbar. **(d)** Anatomical visualization of response duration across regions for each stimulation condition, where darker colors represent longer responses.



Supplemental Figure 8. Relationship between CCEP response latency and distance to the stimulation location. Spearman correlation between the detected onset latency (after spectral smearing due to broadband filtering) and the distance of contact to the site of stimulation.



Supplemental Figure 9. *Gamma Response Profiles of the Cingulate Cortex Effective Connectivity Network.* **(a)** Anatomical profile of average single-pulse electrical stimulation evoked broadband gamma responses concurrent with significant cortico-cortical evoked potentials (CCEPs). Increases in gamma are shown in yellow, decreases are shown in blue. **(b)** Proportions of contacts showing both significant effective connectivity to each cingulate cortex subregion and an evoked broadband gamma response.



Supplemental Figure 10. Comparisons of Stimulation Evoked Gamma Across Conditions and Hemispheres. Gamma responses to stimulation across all contacts between stimulation conditions. Comparisons were also made between hemispheres (left vs right distribution within each condition).

Subj. ID	Age	Length of Stay (Days)	Seizure Onset Zone	Note
BJH024	38	7	multi-focal, deep AR, right temporal pole, unclear onsets, right hippocampal onsets, left hippocampal onset	multifocal epilepticform discharges: bilateral hippocampi, bilateral inferior temporal gyrus, bilateral middle temporal gyrus, bilateral fusiform gyrus, left piriform gyrus, left periform cortex, left perirhinal cortex, left entorhinal cortex and left amygdala
BJH026	40	9	focal seizure ,left ant hippocampus, left posterior hippo, left entorhinal cortex/ anterior parahippocampal gyrus	focal epileptic discharges in left anterior hippocampus (H1-2), left posterior hippocampus (IL1-3), left entorhinal cortex/ anterior parahippocampal gyrus(J1-2), left posterior cingulate cortex (D1-2),left posterior parahippocampal gyrus (K1-5)
BJH030	22	7	focal seizure onsets at right hippocampal body, right hippocampal head, and right amygdala	multi-focal epileptic discharges: bilateral hippocampal region, bilateral amygdala, bilateral parahippocampal gyrus,
BJH031	30	10	right medial temporal lobe	Frequent focal epileptic discharges over bilateral hemisphere, more frequent and prominent in the right hemisphere, and maximal at right posterior hippocampus and right temporal lobe. The background appeared over right subcentral gyrus, dorsal mid insula, anterior thalamus, and right STG, HG, posterior insula, and posterior thalamus.
BJH032	27	9	right hippocampus; left amygdala; left anterior/posterior hippocampus;	Interictal epileptiform discharges in AL1-4, BL1-3, BR1-2, CR4-7; less frequent epileptiform discharges over AR1-2, ER1-4, FR1-4, GR1-8.
BJH035	41	8	left inferior temporal pole, left perirhinal cortex, right anterior temporal pole,	Multifocal epilepticform discharges over left anterior hippocampus, left posterior hippocampus, left amygdala, left perirhinal cortex, left entorhinal cortex, left piriform cortex, right anterior hippocampus, right amygdala, right anterior temporal lobe.
BJH045	22	21	Unidentified	Epileptiform discharges in right supramarginal gyrus, right pre/post central gyrus, right middle frontal gyrus, right superior frontal gyrus, right amygdala, anterior hippocampus.

BJH046	60	14	left posterior MFG, left anterior SFG, left MFG, left posterior SFG	Multi-focal discharges in left MFG/BA46 (HL10-11), left MFG/BA8v (IL 8-9), left anterior SFG/BA9 (ML1-3), and left anterior SFG/BA8, left posterior and anterior SFG/BA9, left hippocampus
BJH050	28	11	right STG, right hippocampus, right parahippocampal gyrus	Frequent epileptiform discharges in right superior temporal gyrus, right anterior/posterior hippocampus and amygdala; Infrequent epileptiform discharges in inferior temporal lobe, entorhinal cortex, and superior temporal lobe.
BJH052	24	8	Left STG, left MTG/STS, left ITG/fusiform gyrus/superior temporal pole, left piriform cortex	Multifocal epileptiform discharges in right anterior hippocampus, left posterior hippocampus, left entorhinal cortex, left STG, left MTG/STS, left ITG/fusiform gyrus, superior temporal pole, left piriform cortex,
BJH053	20	12	right MFG/IFG pars triangularis (lesion), right pars opercularis (lesion), right IFG pars triangularis (lesion)/dorsal anterior insula and right dorsal mid insula.	Focal epileptiform discharges in right MFG/IFG pars triangularis (lesion), right dorsal anterior insula, right IFG pars opercularis/transmantle (lesion)
BJH054	36	4	left MTG, ITG, FuG, STG, HeG, piriform cortex, MFG	Abnormal spike wave discharges in left MTG, ITG, FuG, STG, HeG, piriform cortex, MFG, left anterior and posterior hippocampus, rarely in right amygdala and anterior hippocampus
BJH056	36	10	left parahippocampal gyrus, right collateral sulcus (parahippocampal gyrus/fusiform gyrus junction)	Epileptiform discharges in left/right anterior hippocampus, left/right entorhinal cortex, left/right parahippocampal gyrus, occasionally in left amygdala and posterior hippocampus, and right posterior hippocampus.
BJH058	22	14	Right middle frontal gyrus, right superior frontal gyrus, right anterior hippocampus, right amygdala,	Multi-focal discharges in right middle frontal gyrus, right superior frontal gyrus. Focal epileptiform discharges in right pre-/post-central gyrus, right anterior hippocampus
BJH062	53	6	periventricular cleft and middle/superior frontal gyrus	Frequent epileptiform discharges in left deep cleft, left anterior cleft, occasionally in left superior frontal gyrus, left periventricular cleft, left middle frontal gyrus, left superior frontal gyrus, left middle anterior cleft, left middle anterior cleft, and left superior frontal gyrus

Supplemental Table 1. Patient Demographics.

Region	Condition	Ipsi Difference	Direction Change Ipsi	Ipsi Significance (p-Value)	Ipsi Right Count	Ipsi Left Count
Amygdala	ACC	-0.006327476	No	0.963585434	12	6
Frontal Lobe	ACC	-0.05520696	No	0.005463846	101	76
Hippocampus	ACC	-0.130179078	No	0.021328671	9	7
Insula	ACC	-0.006286412	No	0.759547251	50	31
Operculum	ACC	-0.05167328	Yes	0.133333333	4	2
Orbitofrontal cortex	ACC	0.016996539	No	0.768661087	76	68
Parietal Lobe	ACC				0	4
Somato-Motor Cortex	ACC				0	8
Temporal Lobe	ACC	-0.034851422	No	0.200055286	54	42
Thalamus	ACC				0	0
Amygdala	MCC				13	0
Frontal Lobe	MCC	-0.140119716	No	8.38433E-07	114	189
Hippocampus	MCC	-0.091411734	Yes	0.063636364	9	3
Insula	MCC	-0.122916087	No	5.61487E-05	53	50
Operculum	MCC	-0.058628159	No	0.077064725	43	25
Orbitofrontal cortex	MCC	-0.154607115	Yes	5.5968E-08	42	30
Parietal Lobe	MCC	-0.10378725	No	0.042018858	6	18
Somato-Motor Cortex	MCC	0.030710147	No	0.289312945	81	85
Temporal Lobe	MCC	-0.161233568	No	0.004149393	70	12
Thalamus	MCC				16	0
Amygdala	PCC				12	0
Frontal Lobe	PCC	0.218860357	Yes	3.38945E-05	12	38
Hippocampus	PCC	-0.000962639	Yes	0.066353899	17	4
Insula	PCC	-0.108074975	Yes	0.05994006	9	5
Operculum	PCC	-0.095082488	Yes	0.088888889	8	2
Orbitofrontal cortex	PCC	0.011897799	Yes	0.089348446	23	16
Parietal Lobe	PCC	0.054785631	Yes	0.683516484	4	12

Somato-Motor Cortex	PCC				0	4
Temporal Lobe	PCC	-0.070645464	Yes	0.014640058	136	40
Thalamus	PCC				8	0

Supplemental Table 2. Hemispheric Differences in Gamma Power by Region. Ipsi-Difference: The magnitude difference in gamma response between left and right hemispheres (right vs left). Directional Change Ipsi: Whether there was a difference in the direction of the evoked gamma response (i.e. increase vs decrease). Ipsi Right/Left Count: number of channels included in comparison.

Region	Condition	Avg Gamma Response, Right Hemisphere (Rho)	Avg Gamma Response, Left Hemisphere (Rho)
Amygdala	ACC	0.06908728	0.075414756
Frontal Lobe	ACC	0.03848938	0.09369634
Hippocampus	ACC	0.060897112	0.191076191
Insula	ACC	0.028705081	0.034991492
Operculum	ACC	-0.002980879	0.054654159
Orbitofrontal cortex	ACC	0.142546805	0.125550266
Parietal Lobe	ACC		0.104112489
Somato-Motor Cortex	ACC		-0.026907005
Temporal Lobe	ACC	0.088333385	0.123184807
Thalamus	ACC		
Amygdala	MCC	-4.73255E-05	
Frontal Lobe	MCC	0.044393965	0.184513682
Hippocampus	MCC	-0.015172522	0.106584256
Insula	MCC	0.044163829	0.167079916
Operculum	MCC	0.048140403	0.106768562
Orbitofrontal cortex	MCC	-0.070059829	0.224666945
Parietal Lobe	MCC	0.008044952	0.111832201
Somato-Motor Cortex	MCC	0.064485673	0.033775526
Temporal Lobe	MCC	0.021047709	0.182281277
Thalamus	MCC	0.004035729	

Amygdala	PCC	0.059392992	
Frontal Lobe	PCC	-0.276615081	0.057754724
Hippocampus	PCC	0.055174608	-0.056137247
Insula	PCC	0.003382583	-0.111457558
Operculum	PCC	-0.069548657	0.164631145
Orbitofrontal cortex	PCC	0.037062377	-0.025164578
Parietal Lobe	PCC	-0.055368065	0.000582435
Somato-Motor Cortex	PCC		-0.002570347
Temporal Lobe	PCC	-0.000363677	0.071009141
Thalamus	PCC	-0.026323722	

Supplemental Table 3. Average Gamma Responses by Region and Hemisphere.

Name	Region	Class
Accessory-Basal-nucleus	Accessory Basal Nucleus of the Amygdala	Amygdala
Anterior-amygdaloid-area-AAA	Anterior Amygdaloid Area	Amygdala
Basal-nucleus	Basal Nucleus of the Amygdala	Amygdala
Corticoamygdaloid-transitio	Corticoamygdaloid transition area	Amygdala
Lateral-nucleus	Lateral nucleus of the amygdala	Amygdala
Paralaminar-nucleus	Paralaminar nucleus of the amygdala	Amygdala
G and S cingul-Ant	Anterior part of the cingulate gyrus and sulcus (ACC)	Cingulate cortex
G and S cingul-Mid-Ant	Middle-anterior part of the cingulate gyrus and sulcus (aMCC)	Cingulate cortex
G and S cingul-Mid-Post	Middle-posterior part of the cingulate gyrus and sulcus (pMCC)	Cingulate cortex
G cingul-Post-dorsal	Posterior-dorsal part of the cingulate gyrus (dPCC)	Cingulate cortex
G cingul-Post-ventral	Posterior-ventral part of the cingulate gyrus (vPCC, isthmus of the cingulate gyrus)	Cingulate cortex
S cingul-Marginalis	Marginal branch (or part) of the cingulate sulcus	Cingulate cortex
G and S frontomargin	Fronto-marginal gyrus (of Wernicke) and sulcus	Frontal Lobe
G and S transv frontopol	Transverse frontopolar gyri and sulci	Frontal Lobe
G front inf-Opercular	Opercular part of the inferior frontal gyrus	Frontal Lobe
G front inf-Orbital	Orbital part of the inferior frontal gyrus	Frontal Lobe
G front inf-Triangul	Triangular part of the inferior frontal gyrus	Frontal Lobe
G front middle	Middle frontal gyrus (F2)	Frontal Lobe
G front sup	Superior frontal gyrus (F1)	Frontal Lobe
S front inf	Inferior frontal sulcus	Frontal Lobe
S front middle	Middle frontal sulcus	Frontal Lobe

S front sup	Superior frontal sulcus	Frontal Lobe
CA1-body	Body of the Cornu Ammonis Area 1	Hippocampus
CA1-head	Head of the Cornu Ammonis Area 1	Hippocampus
CA3-head	Head of the Cornu Ammonis Area 3	Hippocampus
CA4-body	Body of the polymorphic layer of the dentate gyrus	Hippocampus
CA4-head	Head of the polymorphic layer of the dentate gyrus	Hippocampus
GC-ML-DG-body	Body of the granule cell and molecular layer of the dentate gyrus	Hippocampus
GC-ML-DG-head	Head of the granule cell and molecular layer of the dentate gyrus	Hippocampus
HATA	Hippocampal-amygdaloid transition area	Hippocampus
hippocampal_fissure	Hippocampal Fissure	Hippocampus
molecular_layer_HP-body	Body of the molecular layer of the hippocampus	Hippocampus
molecular_layer_HP-head	Head of the molecular layer of the hippocampus	Hippocampus
presubiculum-body	Body of the presubiculum	Hippocampus
subiculum-body	Body of the subiculum	Hippocampus
subiculum-head	Head of the subiculum	Hippocampus
G_Ins_lg_and_S_cent_ins	Long insular gyrus and central sulcus of the insula	Insula
G_insular_short	Short insular gyri	Insula
S_circular_insula_ant	Anterior segment of the circular sulcus of the insula	Insula
S_circular_insula_inf	Inferior segment of the circular sulcus of the insula	Insula
S_circular_insula_sup	Superior segment of the circular sulcus of the insula	Insula
G_precentral	Precentral gyrus	Motor Cortex
S_precentral-inf-part	Inferior part of the precentral sulcus	Motor Cortex
S_precentral-sup-part	Superior part of the precentral sulcus	Motor Cortex
G_and_S_occipital_inf	Inferior occipital gyrus (O3) and sulcus	Occipital Lobe
G_cuneus	Cuneus (O6)	Occipital Lobe
G_occipital_middle	Middle occipital gyrus (O2, lateral occipital gyrus)	Occipital Lobe
G_occipital_sup	Superior occipital gyrus (O1)	Occipital Lobe
Pole_occipital	Occipital pole	Occipital Lobe
S_calcarine	Calcarine sulcus	Occipital Lobe
S_collat_transv_post	Posterior transverse collateral sulcus	Occipital Lobe
S_oc_middle_and_Lunatus	Middle occipital sulcus and lunatus sulcus	Occipital Lobe
S_oc_sup_and_transversal	Superior occipital sulcus and transverse occipital sulcus	Occipital Lobe
S_occipital_ant	Anterior occipital sulcus and preoccipital notch (temporo-occipital incisure)	Occipital Lobe
S_parieto_occipital	Parieto-occipital sulcus (or fissure)	Occipital Lobe
G_and_S_subcentral	Subcentral gyrus (central operculum) and sulci	Operculum
Lat_Fis-ant-Horizont	Horizontal ramus of the anterior segment of the lateral sulcus (or fissure)	Operculum
Lat_Fis-ant-Vertical	Vertical ramus of the anterior segment of the lateral sulcus (or fissure)	Operculum

Lat_Fis-post	Posterior ramus (or segment) of the lateral sulcus (or fissure)	Operculum
G_orbital	Orbital gyri	Orbitofrontal cortex
G_rectus	Straight gyrus , Gyrus rectus	Orbitofrontal cortex
S_orbital_lateral	Lateral orbital sulcus	Orbitofrontal cortex
S_orbital_med-olfact	Medial orbital sulcus (olfactory sulcus)	Orbitofrontal cortex
S_orbital-H_Shaped	Orbital sulci (H-shaped sulci)	Orbitofrontal cortex
S_suborbital	Suborbital sulcus (sulcus rostrales, supraorbital sulcus)	Orbitofrontal cortex
G_subcallosal	Subcallosal area, subcallosal gyrus	Other
S_pericallosal	Pericallosal sulcus (S of corpus callosum)	Other
Cerebellum-Cortex		Other
Inf-Lat-Vent		Other
Lateral-Ventricle		Other
Pallidum		Other
Putamen		Other
VentralDC		Other
Unknown		Other
unknown		Other
G_and_S_paracentral	Paracentral lobule and sulcus	Parietal Lobe
G_pariet_inf-Angular	Angular gyrus	Parietal Lobe
G_pariet_inf-Supramar	Supramarginal gyrus	Parietal Lobe
G_parietal_sup	Superior parietal lobule (lateral part of P1)	Parietal Lobe
G_precuneus	Precuneus (medial part of P1)	Parietal Lobe
S_interm_prim-Jensen	Sulcus intermedius primus (of Jensen)	Parietal Lobe
S_intrapariet_and_P_trans	Intraparietal sulcus (interparietal sulcus) and transverse parietal sulci	Parietal Lobe
S_subparietal	Subparietal sulcus	Parietal Lobe
G_postcentral	Postcentral gyrus	Somatosensory Cortex
S_central	Central sulcus (Rolando's fissure)	Somatosensory Cortex
S_postcentral	Postcentral sulcus	Somatosensory Cortex
G_oc-temp_lat-fusifor	Lateral occipito-temporal gyrus (fusiform gyrus, O4-T4)	Temporal Lobe
G_oc-temp_med-Lingual	Lingual gyrus , lingual part of the medial occipito-temporal gyrus , (O5)	Temporal Lobe
G_oc-temp_med-Parahip	Parahippocampal gyrus , parahippocampal part of the medial occipito-temporal gyrus , (T5)	Temporal Lobe
G_temp_sup-G_T_transv	Anterior transverse temporal gyrus (of Heschl)	Temporal Lobe
G_temp_sup-Lateral	Lateral aspect of the superior temporal gyrus	Temporal Lobe
G_temp_sup-Plan_polar	Planum polare of the superior temporal gyrus	Temporal Lobe
G_temp_sup-Plan_tempo	Planum temporale or temporal plane of the superior temporal gyrus	Temporal Lobe

G_temporal_inf	Inferior temporal gyrus (T3)	Temporal Lobe
G_temporal_middle	Middle temporal gyrus (T2)	Temporal Lobe
Pole_temporal	Temporal pole	Temporal Lobe
S_collat_transv_ant	Anterior transverse collateral sulcus	Temporal Lobe
S_oc-temp_lat	Lateral occipito-temporal sulcus	Temporal Lobe
S_oc-temp_med_and_Lingual	Medial occipito-temporal sulcus (collateral sulcus) and lingual sulcus	Temporal Lobe
S_temporal_inf	Inferior temporal sulcus	Temporal Lobe
S_temporal_sup	Superior temporal sulcus (parallel sulcus)	Temporal Lobe
S_temporal_transverse	Transverse temporal sulcus	Temporal Lobe
Pole_temporal	The Temporal pole	Temporal Lobe
Thalamus	Thalamus	Thalamus
Cerebral-White-Matter		White matter
WM-hypointensities		White matter

Supplemental Table 4. Atlas Legend. Region Labels, Region Name, and Region Class definitions used in the present study.

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